CONGRUENT DISSECTIONS OF A TRIANGLE: THE SIX-PIECE TEMPLATE, SCALE THRESHOLDS, AND A CONJECTURED CLASSIFICATION OF THE TILE COUNTS

VICO BONFIOLI

ABSTRACT. A positive integer N is a *tile count* if some triangle can be cut into N pairwise congruent triangles; Erdős asked for the set of tile counts (Problem 634 of `erdosproblems.com`). This problem is not solved here, and nothing in this paper bears on the prime case, which is treated in the earlier versions. The paper is about the *existence* side: which counts are realised, and at which scales. Fix coprime integers $1 \le e < f$ and the tile with sides $(ef,\ f^2-e^2,\ f^2)$; its angles α, β, γ satisfy $3\alpha+2\beta=\pi$. D. Paliy and M. Beeson had shown that the triangles $W_M(e,f)$ and $T_M(e,f)$ built from this tile are tiled for every scale $M \ge f$, and the expectation was that nothing smaller works. (1) **Theorem W**: for $2 \le e \le f-2$, $W_M(e,f)$ is tiled by $M^2(2f^2-e^2)$ tiles, and $T_M(e,f)$ by $M^2(3f^2-e^2)$ tiles, as soon as $Me \ge f$ and $(M-1)(f-e) \ge f$. The proof is a layout of six pieces whose sizes are explicit linear functions of M and three integers; it is formalised in Lean 4 for all (e,f,M). For $e < f/2$ the threshold $Me \ge f$ is sharp, so the scale f is not the threshold in general; this answers the open case $W_4(3,5)$ of the earlier paper and refutes a size conjecture. (2) A collar construction for the isosceles target with base angles α gives, in Lean, the counts $(f^2-1)(2f^2-1)g^2$ for all $g \ge f+1$ $(f \ge 2)$ and gives the tails $465\,g^2$ for $g \ge 5$ and $966\,g^2$, $2485\,g^2$ for $g \ge 2$; the up-set in g was announced earlier by Paliy. (3) Twenty-seven further tile counts are Lean theorems, by macro-piece layouts found by a template search and verified exactly; ninety-seven tilings are shipped as exact data. (4) **Conjecture C** states the set of tile counts as a union of explicit families; its existence half is a Lean theorem for the ranges proved so far, and its exclusion half is open. For the non-squarefree $N \le 10^6$ the census is now 199 140 excluded, 129 108 tile counts, 63 826 undecided; 770 squarefree values stay open. A list of corrections of earlier statements, and a list of what is not proved, are given.

1. INTRODUCTION

1.1. **The problem and what this paper adds.** A triangle Ω (the *target*) is *tiled by N copies of a triangle* `t` (the *tile*) if Ω is the union of N closed triangles with pairwise disjoint interiors, each congruent to `t`; mirror images are allowed and the pieces need not meet edge to edge. N is a *tile count* if some triangle is tiled by N copies of some triangle. The squares, the numbers x^2+y^2, $3k^2$ and $6k^2$ are tile counts for classical reasons [14, 13]. Laczkovich classified the pairs (target, tile) up to the arithmetic of their angles [7, 8]; Beeson, with Roşu, Zhang and Laczkovich, worked through the classes [1, 3, 4, 2, 15, 6]. Paliy published constructions and a proof of the prime case [9, 10]. The earlier versions of this paper [12, 11] contain the exclusion side (the primes $p \equiv 3 \pmod 4$, $p > 3$, are not tile counts; every $N \le 200$ is decided; a census of the non-squarefree $N \le 10^6$). Version 3.0 is about the other side, *which counts are tile counts*, and it is organised around one family of tiles, the one that carries every open question of the census.

Standing of the statements. Every statement carries one of five labels. [PROVED IN LEAN] The statement is a theorem of the accompanying Lean 4 library, checked by the kernel with the standard axioms only (`propext`, `Classical.choice`, `Quot.sound`), no `sorry`; the Lean statement has been compared with the mathematical one *within this work only* (a reading by a person outside the project has not been made). [PROVED BY HAND] A written proof, audited inside the

Date: Version 3.0, 19 October 2026.

project, not formalised. [COMPUTED, EXACT] An exact finite computation, typically an explicit tiling whose pieces are listed in exact arithmetic and verified by an independent exact checker; this is evidence of existence, and for an explicit tiling it is a proof, but it is not a Lean proof. [CLAIMED BY PALIY] A public claim of D. Paliy that we have not re-proved (in some cases we have checked an example). [CONJECTURE] Unproved. The labels are repeated in the tables; where a result has two labels (say, a hand proof and a Lean proof) both are given.

1.2. **Results at a glance.**

result	statement in one line	label and place
Theorem W (§3)	$W_M(e,f)$ and $T_M(e,f)$ tile once $Me \geq f$ and $(M-1)(f-e) \geq f$ $(2 \leq e \leq f-2)$	[PROVED IN LEAN] [PROVED BY HAND]; §3, App. A
Sharpness (Cor. 3.2)	for $2 \leq e < f/2$: $W_M(e,f)$ tiles iff $Me \geq f$	[PROVED IN LEAN]
Collar family (§4)	$(f^2-1)(2f^2-1)g^2$, $g \geq f+1$; tails $465g^2$ $(g \geq 5)$, $966g^2$ and $2485g^2$ $(g \geq 2)$	[PROVED IN LEAN]; collar step with arbitrary height [PROVED BY HAND]
Macro-piece counts (§5)	27 further tile counts (e.g. 3864, 9940, 62125, 279174)	[PROVED IN LEAN], [COMPUTED, EXACT] data
Conjecture C (§6)	the set of tile counts is $\bigcup_k k^2 \cdot \{\text{families}\}$	existence half [PROVED IN LEAN] on the proved ranges; exclusion half [CONJECTURE]
Census (§7)	199 140 / 129 108 / 63 826 (non-squarefree $N \leq 10^6$)	[COMPUTED, EXACT] with label columns
W^{3+} evidence (§8)	all one-leg words refuted for $(4,7)$, $(5,8)$ at $g=3$	[COMPUTED, EXACT] (not formal)

1.3. **Reading guide.** Section 2 defines the tile, the targets and the two tools used throughout (the ladder subdivision and the square closure). Section 3 states Theorem W with a worked example and a proof sketch; the complete proof is Appendix A. Sections 4 and 5 give the other two families of constructions. Section 6 states Conjecture C and what is proved about it. Section 7 gives the census; Section 8 the computer evidence on one open family. Section 9 lists corrections, Section 10 lists what is not proved, and Section 11 gives credits with dates. A reader who wants only the new mathematics can read Sections 2, 3 and 6.

1.4. **Credit and the use of an AI system.** The constructions in Sections 3–5 build on those of D. Paliy (the W/β framework, caps and collars, the α-isosceles up-set, seeds), M. Beeson and G. Roşu (the targets W, T and the methods), B. Herdt, M. Laczkovich, Y. X. Zhang and J. P. Harries; Section 11 says exactly which statement is whose, with dates. These results were obtained with the help of an AI system (Claude, Anthropic); the questions and the directions were the author's, and the proofs have been checked in Lean or by hand as labelled; the provenance table is in the repository README.

2. SETTING

2.1. **The tile.** Throughout, $e < f$ are coprime positive integers and

$$a = ef, \qquad b = f^2 - e^2, \qquad c = f^2, \qquad \kappa = b + c = 2f^2 - e^2.$$

The *tile* $\mathtt{t}(e,f)$ is the triangle with sides a,b,c; α,β,γ are its angles opposite a,b,c. Heron's formula gives $16\,\mathrm{Area}^2 = e^2b^2(4f^2-e^2)$, and one checks $c^2 = a^2 + bc$ and

$$3\alpha + 2\beta = \pi, \qquad \gamma = 2\alpha + \beta, \qquad \cos\alpha = \frac{\kappa}{2c}.$$

These tiles are the tiles of type (T) in the classification of the earlier paper: they are the tiles for which the prime case reduces to a question about $T_1(e,f)$. The angle α/π is irrational; this is not used below. Small examples: $(e,f)=(1,2)$ gives the tile $(2,3,4)$; $(2,3)$ gives $(6,5,9)$; $(2,5)$ gives $(10,21,25)$; $(1,4)$ gives $(4,15,16)$.

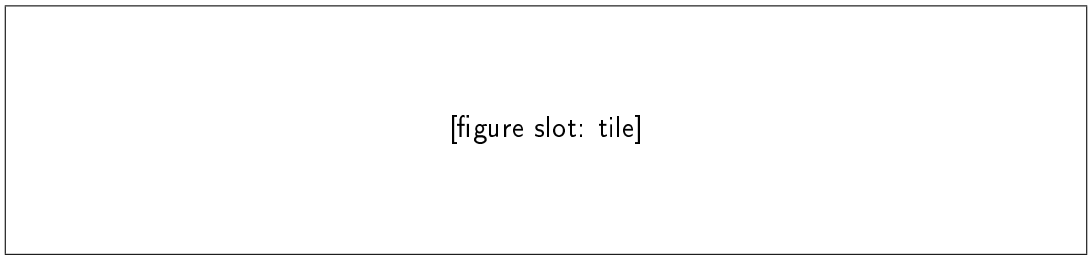


FIGURE 1. The tile $\mathtt{t}(e,f)$ with sides $a=ef$, $b=f^2-e^2$, $c=f^2$ and angles α (opposite a), β (opposite b), $\gamma=2\alpha+\beta$ (opposite c), drawn for $(e,f)=(2,5)$, the tile $(10,21,25)$. The relation $3\alpha+2\beta=\pi$ is what makes the angles of the tile combine into the angles of the targets below.

2.2. **The targets.** Let $M \geq 1$ be an integer. The three families of targets that occur are listed in the table; the last column is the number of tiles that fit by area, which is the only count a tiling of the target can have.

target	sides	angles	area in tiles
$T_M(e,f)$	Mf^3, Mf^3, $Me(3f^2-e^2)$	isosceles; apex 3α, base angles β	$M^2(3f^2-e^2)$
$W_M(e,f)$	Mfb, $Me\kappa$, Mf^3	angles 2α, β, $\alpha+\beta$ (opposite $Me\kappa$, Mfb, Mf^3)	$M^2(2f^2-e^2)$
$B_g(e,f)$	gbc, gbc, $gb\kappa$	isosceles; base angles α, apex $\alpha+2\beta$	$b\kappa\, g^2$

(For W_M, the side opposite the angle 2α is $Me\kappa$, opposite β is Mfb, and opposite $\alpha+\beta$ is Mf^3.) T_M and W_M are the targets of Beeson–Roşu and Paliy ("base-β" targets); B_g is the "base-α" target (Paliy's α-isosceles). We call a target *tiled* if it is tiled by copies of the tile $\mathtt{t}(e,f)$ of the same (e,f). $W_M(e,f)$ and $T_M(e,f)$ are related by gluing: a tile at scale Mf glued to the side Mfb of W_M gives T_M (§3.3).

2.3. **Two tools.**

Proposition 2.1 (Ladder subdivision; [PROVED IN LEAN]). *Let $s \geq 1$ be an integer. A triangle similar to $\mathtt{t}$ with sides sa, sb, sc is tiled by s^2 copies of $\mathtt{t}$. A parallelogram with sides sa and uc enclosing the angle β is tiled by $2su$ copies of $\mathtt{t}$ (cut each unit cell along its diagonal of length b).*

Lean: `Subdivision.ladderDissection`, `Ladder.ladder`; the earlier paper calls this `thm:ladder`. A *trapezoid* below means a triangle of scale s_1 minus a corner triangle of scale $s_2 < s_1$ of the same shape; it contains $s_1^2 - s_2^2$ tiles.

Proposition 2.2 (Square closure; [PROVED IN LEAN]). *If N is a tile count then so is k^2N for every integer $k \geq 1$.*

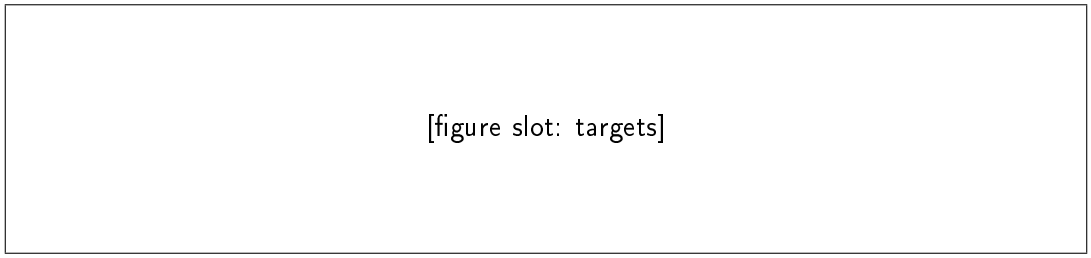


FIGURE 2. The three targets for $(e, f) = (2, 5)$ and $M = 1$, drawn to scale: $W_1(2, 5)$ (sides $105, 92, 125$), $T_1(2, 5)$ (legs 125, base 142), and $B_1(2, 5)$ (legs 525, base 966). The tile $(10, 21, 25)$ is drawn inside each for comparison. The angles are marked; they are built from α, β by $3\alpha + 2\beta = \pi$.

Lean: `TileCountFamilies.tileCount_scale`. (Cut every tile of a tiling of Ω into k^2 triangles similar to it, with ratio $1/k$; the k^2N pieces are congruent.)

Remark 2.3. In words: scaling the whole figure by k multiplies the number of tiles by k^2. Because of Proposition 2.2 a tile count is naturally written $N = k^2 n$, and thresholds below are thresholds in the *scale of the target relative to the tile*, not in k.

3. Theorem W: the six-piece template

3.1. Statement.

Recall $a = ef$, $b = f^2 - e^2$, $c = f^2$, $\kappa = 2f^2 - e^2$, and put $g = f - e$.

Theorem 3.1 (Theorem W; [proved in Lean] and [proved by hand]). *Let e, f be coprime integers with $2 \le e \le f - 2$, and let $M \ge 1$ be an integer with*

$$\textit{(H1)} \quad Me \ge f \qquad \textit{and} \qquad \textit{(H2)} \quad (M-1)(f-e) \ge f.$$

Then $W_M(e, f)$ is tiled by $M^2(2f^2 - e^2)$ copies of $\mathsf{t}(e, f)$ and $T_M(e, f)$ is tiled by $M^2(3f^2 - e^2)$ copies of $\mathsf{t}(e, f)$.

Lean: `Draft.ThmW.thmW`: for natural numbers `e f M` with `Nat.Coprime e f`, `2 <= e`, `e+2 <= f`, `f <= M*e`, `f <= (M-1)*(f-e)`, the statement `WTilable e f M ∧ TileCount ((2*f^2-e^2)*M^2) ∧ TileCount (M^2*(3*f^2-e^2))`. Here `WTilable e f M` says that there is a dissection into congruent triangles, each congruent to the triangle with sides ef, $f^2 - e^2$, f^2, of a triangle with sides $Mf(f^2 - e^2)$, $Me(2f^2 - e^2)$, Mf^3; the two `TileCount` conjuncts say only that the number is a tile count (the shape is pinned by the first conjunct). The module is `Draft/ThmW.lean` with its `ThmW*` helper modules; the aggregate is `Erdos634.AllThmW`; the axioms are `propext`, `Classical.choice`, `Quot.sound`. The statement of the theorem was compared with the hand statement in an audit inside the project (verdict: match, with a note about credits only). The hand proof, given in full in Appendix A, was audited separately by an adversarial reader and by exact computation (Section 3.6).

Why it matters. Before this theorem, $W_M(e, f)$ and $T_M(e, f)$ were known to be tiled for $M \ge f$ (Paliy, Beeson), and the working expectation was that no smaller M works for $e \ge 2$; the only smaller scales known were the cases $e = 1$ and $e = f - 1$, where the threshold is indeed f. Theorem W shows that for $e \ge 2$ the scale f is *not* the threshold: for $(e, f) = (2, 5)$ the scale 3 works instead of 5, and for $e < f/2$ the new threshold is exactly the proved lower bound (Corollary 3.2).

Put $M^*(e, f) = \max\big(\lceil f/e \rceil,\ \lceil f/(f-e) \rceil + 1\big)$; the hypotheses (H1), (H2) say $M \ge M^*(e, f)$.

(e,f)	M^*	first tiled target	number of tiles	previously known threshold
$(2,5)$	3	$W_3(2,5)$	414	5
$(3,5)$	4	$W_4(3,5)$	656	5
$(3,7)$	3	$W_3(3,7)$	801	7
$(2,11)$	6	$W_6(2,11)$	8568	11
$(9,11)$	7	$W_7(9,11)$	7889	11

The thirteen values $M < f$ with $4f < 9e$ that were found first by a computer search are listed in Appendix B; they are instances of the theorem.

3.2. **Consequences.**

Corollary 3.2 (Sharpness for $e < f/2$; [PROVED IN LEAN]). *Let $2 \le e < f/2$, $\gcd(e,f) = 1$. Then $W_M(e,f)$ is tiled by copies of $\mathtt{t}(e,f)$ if and only if $Me \ge f$.*

Proof. If $Me < f$ there is no tiling: this is the earlier [PROVED IN LEAN] corollary `cor:ebelow`(a) (Lean `Draft.SideRunG.ebelow_a`, in the release paper [11]). Conversely, $e < f/2$ gives $f - e > f/2$, so $f/(f-e) < 2$ and $\lceil f/(f-e)\rceil + 1 = 3$, while $f/e > 2$ gives $\lceil f/e\rceil \ge 3$; hence $M^*(e,f) = \lceil f/e\rceil$ and (H1) implies (H2). Apply Theorem 3.1. □

Corollary 3.3 (Answer to an open case; [PROVED IN LEAN]). *$W_4(3,5)$, with sides 320, 492, 500, is tiled by 656 copies of $(15,16,25)$. A tiling is shipped as exact data (file `tiling_656_W_3_5_M4`).*

The earlier paper left the scales $M < f$ open apart from a few instances, and recorded the pattern "every case decided so far is tiled exactly when $M \ge f$" ([11], Section on scales). Corollary 3.3 is a counterexample to that pattern: $M = 4 < 5 = f$. It also refutes a size conjecture of the working notes (a target not similar to the tile has longest side at least c^2): here $500 < 625 = c^2$ (see Section 9).

3.3. **The T half.** Glue to the side Mfb of $W_M(e,f)$ one copy of the tile at scale Mf (sides Mef^2, Mfb, Mf^3) with its side Mfb on that side and its far vertex Z on the line AB, beyond B (so $BZ = Mef^2$); the result is the isosceles triangle with legs Mf^3 and base $Me(3f^2 - e^2)$, that is $T_M(e,f)$, and it has $M^2(2f^2 - e^2) + M^2f^2 = M^2(3f^2 - e^2)$ tiles. This is the earlier `prop:layerall`(a) (Lean `CollarG.betaTilable_of_wTilable`); the Lean statement of Theorem W includes it (`DebtB.counts_of_wTilable` needs only the weaker hypothesis $M(f-e) \ge f$ on the T side). The converse (a tiling of T_M gives one of W_M) is not known; the proved exclusions for T_M are weaker than for W_M (only $2Me < f$, and $e = 1$, $2M \le f$), so the T column of the census has more open scales (Section 10).

3.4. **The six pieces, in words, and a worked example. Idea.** The target $W_M(e,f)$ has a vertex O with angle 2α and two other vertices A (angle β) and B (angle $\alpha+\beta$). Let P be the point of the side AB with $OP = OB$; then OP bisects the angle at O, and the triangle OBP is isosceles with apex angle α. The template cuts W_M into six pieces, each of which is tiled by Proposition 2.1:

piece	kind	scales (integers)	tiles
π_0	trapezoid	triangle OAP of scale fM minus a corner triangle of scale et	$f^2M^2 - e^2t^2$
π_1	triangle	scale $v = fM - er$	v^2
π_2	triangle	scale ft	f^2t^2
π_3	trapezoid	scales $s_1 = f(r-t)$ and $s_2 = fr - eM$	$s_1^2 - s_2^2$
π_4	parallelogram	sides as_1 and cx, $x = ft - eq$, angle β	$2s_1x$
π_5	trapezoid	scales $u_1 = fq - et$ and $u_2 = fq - er$	$u_1^2 - u_2^2$

with four free integers M, t, r, q. The six tile counts add up to $M^2(2f^2 - e^2)$ identically in e, f, M, t, r, q (a polynomial identity), so what remains is to place the pieces: all vertices are given by vector sums with coefficients linear in M, t, r, q, and "the pieces lie in the target and do not overlap" becomes a list of 26 inequalities between linear and quadratic expressions in e, f, M, t, r, q (Appendix A, Lemma B). The proof chooses

$$n = \left\lfloor \frac{eM}{f} \right\rfloor, \quad \rho = eM - fn, \quad \delta = \begin{cases} 1 & \rho = 0 \\ 0 & \rho > 0 \end{cases}, \qquad t = n - \delta, \quad r = n + 1, \quad q = n,$$

and shows that under (H1), (H2) every inequality holds (Lemma C).

Example 3.4 ($W_3(2,5)$ with 414 tiles)**.** $(e, f, M) = (2, 5, 3)$: $a = 10$, $b = 21$, $c = 25$, $\kappa = 46$, and $W_3(2,5)$ has sides 315, 276, 375. Here $n = \lfloor 6/5 \rfloor = 1$, $\rho = 1$, $\delta = 0$, so $(t, r, q) = (1, 2, 1)$, and

	scales	tiles		scales	tiles
π_0	15 minus 2	$225 - 4 = 221$	π_3	5 minus 4	$25 - 16 = 9$
π_1	$v = 11$	121	π_4	$as_1 \times cx$, $s_1 = 5$, $x = 3$	$2 \cdot 5 \cdot 3 = 30$
π_2	5	25	π_5	3 minus 1	$9 - 1 = 8$

The total is $221 + 121 + 25 + 9 + 30 + 8 = 414 = 9 \cdot 46$. A full list of the 414 tiles is shipped (file `tiling_414_W_2_5_M3`) and has been verified by an independent exact checker.

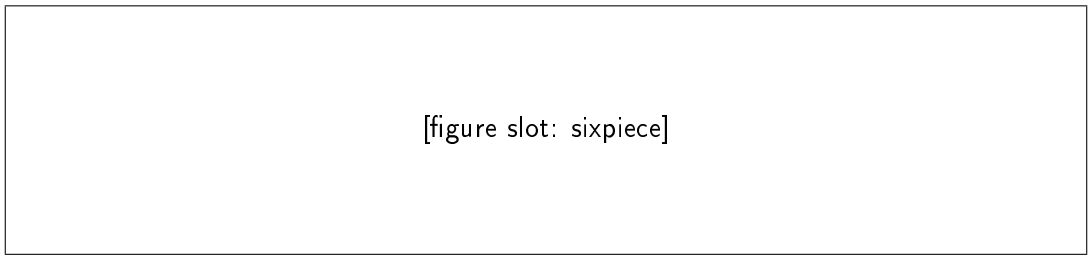


FIGURE 3. The six-piece layout of Theorem W for $W_3(2,5)$, a triangle with sides 315, 276, 375 tiled by 414 copies of $(10, 21, 25)$. The pieces $\pi_0, \ldots, \pi_5$ are shaded differently and labelled with their kind and scales as in Example 3.4: π_0 is the large trapezoid at the vertex O of angle 2α (triangle OAP of scale 15 minus a corner of scale 2), π_1 the triangle of scale 11 on the side OB, π_2 the triangle of scale 5 on the side AB, π_3 and π_5 the two small trapezoids, π_4 the parallelogram of 30 tiles. The point P on AB with $OP = OB$ is marked.

3.5. **Proof sketch.** (1) *Shapes.* Each piece is the stated union of tiles when its scales are positive integers: the triangles by SAS/SSS, the trapezoids because the corner triangle is cut off by a line parallel to a side, the parallelogram by Proposition 2.1. The shape claims are identities of Eisenstein-type numbers $x + i\,w\,y$ with $w = \sqrt{4f^2 - e^2}$ and use only $3\alpha + 2\beta = \pi$ (equivalently $\varepsilon^3\eta^2 = -1$ for $\varepsilon = e^{i\alpha}$, $\eta = e^{i\beta}$) and the closure relation of the tile. (2) *Placement.* The pieces are convex and counter-clockwise; they lie in the target iff their vertices do, and two pieces have disjoint interiors if one edge line of one piece separates it from the other. Together with the area identity this gives a tiling. Each needed sign is a product of positive constants and of factors from a fixed list of 26 linear or quadratic expressions in e, f, M, t, r, q. (3) *Choice of parameters.* With n, ρ, δ, t, r, q as above and $s = M - n$ (so $gM = fs - \rho$ for $g = f - e$), every factor is positive. The only places where (H1) and (H2) are used are: $n \geq 1$ and $gn \geq e + 1$ (from

(H1) and (H2) and coprimality), and $s \geq 2$ (from (H2)); the decisive inequality is the identity (LB) $= s\,\Psi - \kappa(\rho + e)$ with $\Psi = f^3 + 2ef^2 - e^2 f - e^3 > 0$, which is positive because $s \geq 2$ and $\rho \leq f - 1$. (4) *Lean.* Items (1)–(3) are formalised as separate modules (point lemmas, shape lemmas, one sign lemma per factor, the parameter lemma, the packing); the packing uses the general theorem that a finite family of convex pieces, each tiled, with disjoint interiors and total area equal to the area of the target, tiles it.

3.6. **Evidence besides the proof.** [COMPUTED, EXACT] For *every* coprime pair $2 \leq e \leq f - 2$ with $f \leq 300$ (26 800 pairs), an exact integer program checked the scale positivity, the SSS shapes, the trapezoid ratios, convexity, containment, the area sum and a separating edge for all 15 pairs of pieces, for M from M^* to $M^* + 7$ and for $M \in \{f, f + 1, 2f - 1, 2f, 3f\}$: 348 350 cases, no failure. Negative controls: for $M = M^* - 1$ the canonical parameters fail for every pair tried. A second, independent checker (own code, a different arithmetic) repeated this for all $f \leq 120$ (706 489 cases) and for random $f \leq 400$. Thirty-eight explicit tilings were expanded to the full list of tiles and verified by an exact checker (congruence, pairwise disjointness, area, the hull is a triangle), among them $W_3(2, 5)$, $W_3(3, 7)$, $W_5(2, 9)$ (3950 tiles), $W_8(11, 13)$ (13 888 tiles), $W_4(11, 43)$ (57 232 tiles) and $T_3(2, 5)$, $T_6(3, 16)$, $T_7(24, 29)$ (95 403 tiles). No instance of the theorem contradicts a proved exclusion (checked for all $e, f \leq 200$: 4 786 740 admissible triples), and none of the 197 recorded exclusions of the census registry meets the family.

3.7. **What stays open.** For $e < f/2$ nothing: Corollary 3.2. For $e > f/2$ the *no* side of (H2) is open: the proved exclusion is only $M(f - e) \leq e$ (earlier `cor:thickw`(a)), so for each row (e, f) exactly the (at most two) scales M with $(M - 1)(f - e) < f < (M + 1)(f - e)$ are undecided; the conjecture is that W_M is tiled iff $(M - 1)(f - e) \geq f$. The members $e = 1$ and $e = f - 1$ are outside the theorem; for both the threshold is known to be $M = f$ (`cor:below`, `cor:thickw`(b)). For T_M the open scales are those with $2Me \geq f$ and $(M - 1)(f - e) < f$, the first being $T_3(1, 4)$ with $e = 1$ and, for $e \geq 2$, for example $T_2(3, 5)$ and $T_3(3, 5)$.

4. The collar family and the base-α rows

This section is about the third target of Section 2.2, the isosceles triangle $B_g(e, f)$ with base angles α, legs gbc, base $gb\kappa$, and $b\kappa g^2$ tiles. Write $N_0(e, f) = b\kappa = (f^2 - e^2)(2f^2 - e^2)$, so that a tiling of $B_g(e, f)$ has $N_0(e, f)g^2$ tiles; *the row* (e, f) is the set of g for which $B_g(e, f)$ is tiled. The first values: $N_0(1, 2) = 21$, $N_0(2, 3) = 70$, $N_0(1, 4) = 465$, $N_0(2, 5) = 966$, $N_0(1, 6) = 2485$. The row is non-empty in every case examined, and it is closed under addition (Paliy, [CLAIMED BY PALIY]; his construction), so it is, if his claim holds, a numerical semigroup; what is open is its smallest element and its gaps.

4.1. **A priority statement.** D. Paliy stated on 29 September 2026 [10] that the tile counts of the form $b\kappa K^2$ are exactly the counts of tilings of the targets B_K and that, for each tile, every K beyond an explicit threshold K_0 is realised (the "α up-set"); the seeds 84 and 189 of the row $(1, 2)$ are in Beeson's work (the tilings of 84 and 189 in his Figures 28 and 29; 189 is due to Harries). **The up-set and the counts are Paliy's.** What is ours in this section is an explicit construction of the layers (the *collar*), a parametric seed for the whole family $e = 1$, the explicit thresholds that follow, and their Lean proofs; the construction is independent of his, and the thresholds below are lower than the ones obtained from his formula as we read it for the rows $(1, 4)$, $(1, 6)$ and $(2, 5)$ of Theorem 4.3 (we have not asked him).

4.2. **The collar step.**

Theorem 4.1 (Collar step; [PROVED BY HAND], with a [PROVED IN LEAN] one-layer case)**.** *Let $e < f$ be coprime, $a = ef$, $b = f^2 - e^2$, $\kappa = 2f^2 - e^2$, and suppose $B_{g_0}(e, f)$ is tiled. Choose $s \in [0, a)$ with $bs \equiv b\kappa g_0 \pmod a$ (it exists since $\gcd(a, b) = 1$), put $K = (b\kappa g_0 - bs)/a$ and $h_{\max} = \lfloor K/a \rfloor$. Then for every integer h with $1 \le h \le h_{\max}$ the triangle $B_{g_0+h}(e, f)$ is tiled.*

Construction. B_{g_0+h} is a homothetic copy of B_{g_0} at the apex plus a trapezoidal strip along the base; the strip is cut into three pieces by two lines, each tiled by Proposition 2.1: a triangle of scale ch, the difference of two triangles $\Delta(K) \setminus \Delta(K - ah)$, and the difference $\Delta(s + bh) \setminus \Delta(s)$. The identity that makes the cut edges coincide is $c^2 h + a(K - ah) = b\kappa(g_0 + h) - b(s + bh)$, which uses only $c^2 = a^2 + bc$ and $\gamma = 2\alpha + \beta$; the range $h \le K/a$ is exactly the condition $K - ah \ge 0$. *Status.* The hand proof was audited independently (verdict: sound; checked exactly on 17 853 instances, all coprime $e < f < 40$ with several (g_0, h) each, and no failure). [PROVED IN LEAN] The case $h = 1$, for every row with $ef \le f^2 - e^2$, is Lean `Draft.CollarFam.layer_step` (real-level core `Draft.CollarLayer.layer_core`); arbitrary h is hand. The corollary "g_0 in the row implies every $g \ge g_0$ is in the row" needs $h_{\max}(g_0) \ge g_0 - 1$ together with $h_{\max}(x) \ge 1$ for $x \ge 2g_0 - 1$ (the first version of this claim in our notes omitted the second condition and was corrected by the audit); it holds for the rows below.

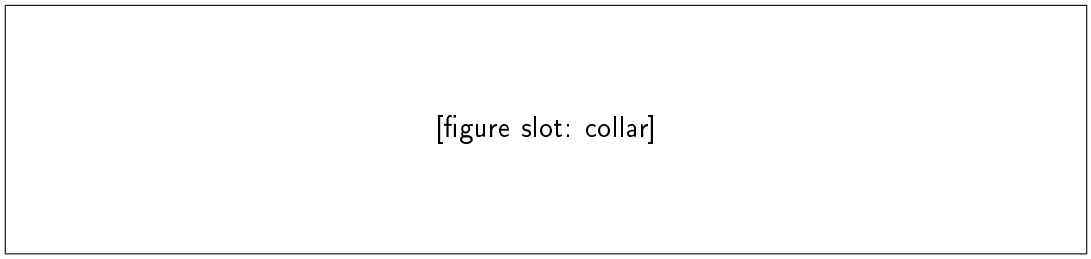


FIGURE 4. The collar: $B_{g_0+h}(1, 4)$ with $g_0 = 5$, $h = 3$ (the tiling of $465 \cdot 8^2 = 29\,760$ tiles). The inner triangle is the homothetic copy of $B_5(1, 4)$ (tiled by 11 625 tiles, itself a six-piece layout); the strip below it is cut into the three pieces of Theorem 4.1, labelled with their scales. Each piece is a ladder subdivision.

4.3. **The seed for $e = 1$.**

Theorem 4.2 (Seed and up-set for $e = 1$; [PROVED IN LEAN])**.** *For every $f \ge 2$ and every $g \ge f + 1$, $(f^2 - 1)(2f^2 - 1)\, g^2$ is a tile count; the tiling is of $B_g(1, f)$ by copies of $(f, f^2 - 1, f^2)$.*

Lean: `Draft.CollarUp.tileCount_e1_up`. The seed $g = f + 1$ is a six-piece layout whose pieces are polynomial in f (generated and checked symbolically; module `CollarSeed`, 125 lemmas), and the layer step then gives all $g \ge f + 1$. For $f = 2, 3, 4, 6$ the seed counts are 189, 2176, 11 625, 121 765; the first is the 189 of Beeson and Harries, the third and fourth were found by a template search in release v1.16.

Why it matters. The sporadic-looking counts 189, 11 625, 121 765 of earlier releases are the first members of one family, and the count $(f^2 - 1)(2f^2 - 1)g^2$ is a tile count for all $f \ge 2$ at scales that no longer depend on a search.

4.4. **Lowered tails for three rows.**

Theorem 4.3 (Tails; [PROVED IN LEAN])**.** *(a) $465\, g^2$ is a tile count for every $g \ge 5$ (row $(1, 4)$, tile $(4, 15, 16)$). (b) $2485\, g^2$ is a tile count for every $g \ge 2$ (row $(1, 6)$, tile $(6, 35, 36)$). (c) $966\, g^2$ is a tile count for every $g \ge 2$ (row $(2, 5)$, tile $(10, 21, 25)$).*

Lean: (a) `Draft.CollarFam.tileCount_465_sq`, from the seed 11 625 and the layer step. (b) The tail $g \geq 5$ is `Draft.CollarFam5.tileCount_2485_sq5`; $g = 2, 3$ are 9940 and 22 365 (Section 5), and $g = 4$ is $4 \cdot 9940$ by Proposition 2.2. (c) The tail $g \geq 17$ is `Draft.CollarFam17.tileCount_966_sq17`; $g = 2, \ldots, 8$ and $10, \ldots, 18$ are the macro-piece counts $3864, \ldots, 312\,984$ of Section 5 (sixteen values), and $g = 9$ is $9 \cdot 8694 = 78\,246$ (`Draft.AllMacro.tileCount_78246`). The squarefree cores 465, 966 and 2485 themselves ($g = 1$) are not tile counts known to us; they are among the open squarefree values.

Remark 4.4. In our working notes we had recorded the pattern "the smallest scale of a row is $f + 1$ for even f". (c) and (b) refute it: the rows $(2, 5)$ and $(1, 6)$ contain $g = 2$. For $(1, 4)$ the values $g = 2, 3, 4$ (1860, 4185, 7440) are not known to be tile counts or non-tile counts. Paliy's threshold, read from his formula, is $K_0 = 17$ for $(2, 5)$ (equal to our tail in (c) before the macro pieces fill $g = 2, \ldots, 16$), 6 for $(1, 4)$ and 8 for $(1, 6)$ (we did not check these values with him).

5. Macro-piece tile counts

5.1. **The idea.** In the six-piece layout of Section 3 every piece is tiled by the ladder subdivision, that is, by a tiling of a *triangle similar to the tile*. A *macro piece* is a piece that is a triangle similar, with integer ratio k, to the *target* of a tiling already known: such a piece is tiled by $k^2 n$ tiles if the known tiling has n tiles (Lean: `Draft.MacroPiece.macroTFam_gen`, which transports a known dissection by the ladder theorem and congruence of triangles with equal sides). Macro pieces were used for a long time by hand (Laczkovich's macro-dissections, Herdt's 1125; Section 6 of the version-2.2 paper), and they enlarge the set of layouts for which the area equation has integer solutions. The layouts are found by a computer search over the combinatorial type of the layout and the integer scales of its pieces, with the equations solved exactly; a layout is then expanded to the full list of tiles, verified by an exact checker, and finally turned into a Lean module, whose proof is the same area-and-containment argument as in Theorem W with the numbers of that layout. The search itself is not part of this release; what is shipped are the results: the exact tilings (ninety-seven files, Appendix B) and the Lean theorems.

5.2. **The counts.**

Theorem 5.1 (Macro-piece tile counts; [proved in Lean], [computed, exact] data)**.** *Each of the following is a tile count, as the statement* `TileCount` N *of a Lean theorem:*

values N	*form*	*tile*	*what is tiled*
375, 540, 735	$15m^2$, $m = 5, 6, 7$	$(3, 5, 7)$	*equilateral triangles of side* $75, 90, 105$
825	$33 \cdot 5^2$	$(3, 5, 7)$	*isosceles, sides* $105, 105, 165$
630, 1120, 1750	$70\,g^2$, $g = 3, 4, 5$	$(6, 5, 9)$, *row* $(2, 3)$	$B_g(2, 3)$
3864, 8694, 15 456, 24 150, 34 776, 47 334, 61 824	$966\,g^2$, $g = 2, \ldots, 8$	$(10, 21, 25)$, *row* $(2, 5)$	$B_g(2, 5)$
96 600, 116 886, 139 104, 163 254, 189 336, 217 350, 247 296, 279 174, 312 984	$966\,g^2$, $g = 10, \ldots, 18$	$(10, 21, 25)$	$B_g(2, 5)$
9940, 22 365, 62 125	$2485\,g^2$, $g = 2, 3, 5$	$(6, 35, 36)$, *row* $(1, 6)$	$B_g(1, 6)$

That is 26 *values, and a twenty-seventh,* $78\,246 = 9 \cdot 8694$*, by Proposition 2.2.*

Lean: one module `Draft/TM`N (or `Draft/T62125`) per value, with `theorem tileCount_`N `: TileCount` N; aggregate `Draft.AllMacro`. Each module fixes the layout, proves the shapes

and the containment by the area-and-sign method, and does not use a tile list; the standard axioms only. The exact tilings of all of them are in `data/witnesses/v3_exact` (for $N = 375$, `tiling_375_macro_5_3_7_g1`); they have been verified by an exact checker independently of the Lean proofs.

Why it matters. Together with Theorem 4.3 these counts show that in two rows, $(2, 5)$ and $(1, 6)$, *every* scale $g \geq 2$ is realised; the squarefree cores of those rows are the only open members. The $15m^2$ values for $m = 5, 6, 7$ are the seeds that Paliy used ([CLAIMED BY PALIY]) for his claim that $15m^2$ is a tile count for every $m \geq 4$.

For the equilateral counts $15m^2$: $m = 4$ (240) was already a Lean theorem, and so was every m in the semigroup generated by 4 and 9 (earlier release). The Lean theorem `Draft.EqSemigroup.eqD_step` says that equilateral tilings of sides $15m_1$ and $15m_2$ by copies of $(3, 5, 7)$ give one of side $15(m_1 + m_2)$; the seeds $4, 5, 6, 7$ would therefore give every $m \geq 4$ (this is Paliy's claim, [CLAIMED BY PALIY]). The new Lean theorems for $m = 5, 6, 7$ state only `TileCount` and are not yet instances of the equilateral predicate used by `eqD_step`, so "$15m^2$ for all $m \geq 4$" is *not* a Lean theorem here; the exact tilings (sides $75, 90, 105$) are verified, and $15m^2$ for the individual $m = 5, 6, 7$ is Lean.

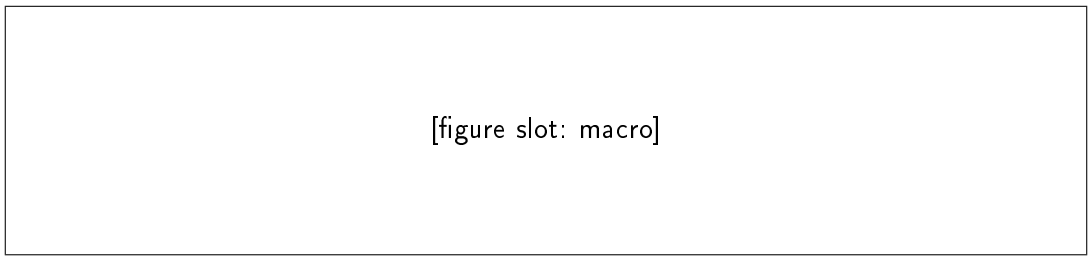


FIGURE 5. Composition of a macro-piece tiling: a template layout of $B_g(2, 5)$ with $g = 6$ ($34\,776 = 966 \cdot 36$ tiles) whose pieces are ladder triangles, parallelograms, trapezoids and one *macro piece*, a triangle similar with integer ratio to the target of the already-known tiling of $966 \cdot g'^2$ tiles; the macro piece is drawn with the known tiling inside it.

5.3. **Other counts shipped in this release.** Twelve counts that are squares times earlier template counts ($46\,500$, $104\,625$, $267\,840$, $290\,625$, $418\,500$, $476\,160$, $487\,060$, $569\,625$, $602\,640$, $671\,460$, $744\,000$, $941\,625$; module `Draft/TemplateSquares`) are Lean theorems by Proposition 2.2; they were already tile counts mathematically and had been omitted by the census bookkeeping. The thirteen tilings of W_M with $M < f$ of Appendix B are also Lean theorems (`Draft/AllWBelow`), as are their upward closures (`Draft/AllDebtB`); they are subsumed by Theorem W but were proved first. 4830 is a tile count: Paliy's certificate (4830 copies of $(24, 11, 31)$ tile the triangle $(961, 1155, 1426)$) has been converted and verified exactly ([COMPUTED, EXACT]; the construction is Paliy's, [CLAIMED BY PALIY]; his family of mixed-corner layouts gives further values that we have not verified).

6. CONJECTURE C: A CANDIDATE CLASSIFICATION OF THE TILE COUNTS

The earlier papers give the exclusion side and a growing list of tile counts, but no statement of what the set of tile counts *is*. This section states one, as a conjecture, and says exactly what part of it is proved. **It is not claimed to be true.** It is an organising map: a new tile count is either in a family (no news), or extends a family's range, or is a counterexample to the conjecture.

6.1. **The statement.** Recall Proposition 2.2: every family below is understood closed under $n \mapsto k^2 n$.

Conjecture 6.1 (Conjecture C; [CONJECTURE]). *A positive integer N is a tile count if and only if $N = k^2 n$ for some integer $k \geq 1$ and some core n in one of the families $F_0, \ldots, F_8$ of Table 1, with the parameter ranges stated there.*

	cores n	tile, target	range proved (label) / credit
F_0	1, x^2+y^2 $(x,y \geq 1)$, 3, 6	similar tiles, rep-tiles	all [PROVED IN LEAN] / Snover–Waiveris–Williams, Laczkovich
F_1	$2f^2 - e^2$ and $3f^2 - e^2$, at scale M: $M^2(2f^2 - e^2)$, $M^2(3f^2 - e^2)$	$\mathtt{t}(e,f)$; W_M, T_M	$e = 1$ or $e = f-1$: $M \geq f$. $2 \leq e \leq f-2$: $M \geq M^*(e,f)$ (Thm 3.1). [PROVED IN LEAN] / Paliy, Beeson ($M \geq f$); ours (M^*)
F_2	$(2q^2 - p^2)(3q^2 - p^2)$, $0 < p < q$	five-piece layouts	all [PROVED IN LEAN] / Paliy
F_3	$(2a+b)(a+b)$, $3(a+b)(a+2b)$, $(a+2b)(2a+b)$ and Zhang's five forms $\times m^2$; $15m^2$, $33m^2$	tiles with an angle $2\pi/3$, $c^2 = a^2 + ab + b^2$; $(3,5,7)$	ranges as in the Lean predicates; $15m^2$: $m \in \langle 4, 9\rangle$ and $m = 5, 6, 7$ [PROVED IN LEAN] / Paliy, Zhang, Herdt. $15m^2$ for all $m \geq 4$: [CLAIMED BY PALIY]
F_4	88, 143 (sporadic)	tile $(3,5,7)$ on the triangles $(21,55,56)$ (88) and $(39,49,55)$ (143)	[PROVED IN LEAN] / Harries (88); ours (143)
F_5	$(f^2-e^2)(2f^2-e^2)\,g^2$, $g \in G(e,f)$	$\mathtt{t}(e,f)$; $B_g(e,f)$	$e = 1$: $g \geq f+1$; $(1,4)$: $g \geq 5$; $(1,6), (2,5)$: $g \geq 2$; $(2,3)$: $g = 3, 4, 5$ and more by Thm 4.1 [PROVED IN LEAN] / [PROVED BY HAND]; Paliy: eventual up-set in every row, [CLAIMED BY PALIY]
F_6	$b\,(v(D+\tau v))^2$ with two Diophantine conditions; 1125, 19 600, 130 977	$\gamma = 2\alpha$, tile $(u^2, v^2 - u^2, uv)$	Herdt's family and its uniform version [PROVED IN LEAN] / Herdt, Laczkovich
F_7	4830	$(24, 11, 31)$	[COMPUTED, EXACT] / Paliy, [CLAIMED BY PALIY] (mixed-corner family)
F_8	bt^2, $t \in S(e,f)$	$\mathtt{t}(e,f)$; θ-isosceles target (legs tfb, base teb, base angles $\alpha+\beta$, bt^2 tiles)	[CLAIMED BY PALIY] / Paliy (semigroup $S(e,f)$, $te \geq 2f$ necessary [PROVED IN LEAN])

TABLE 1. The families of Conjecture C and what is proved about each. $\langle 4, 9\rangle$ is the set of non-negative combinations of 4 and 9. In F_1 the ranges are those of Theorem 3.1 and of the earlier theorem for $M \geq f$; for the members $e = 1$ and $e = f - 1$ the threshold f is proved sharp for W_M.

Remarks. (i) $84 = 21 \cdot 2^2$ and $189 = 21 \cdot 3^2$ are not sporadic: they are the members $g = 2, 3$ of the row $(1, 2)$ of F_5 (earlier lists had them in F_4); $77 \in F_2$; 183 is *not* a tile count (it is among the excluded values ≤ 200; an earlier draft of this section listed it by a misreading, see Section 9). (ii) Theorem 3.1 is stated for $2 \leq e \leq f - 2$; F_1 in Table 1 includes the two edge members with their threshold f. (iii) For F_6 the earlier papers state two conjectures about its exclusion side, V ($v \mid g$ is necessary) and A' (a size bound); Section 8 gives evidence for them.

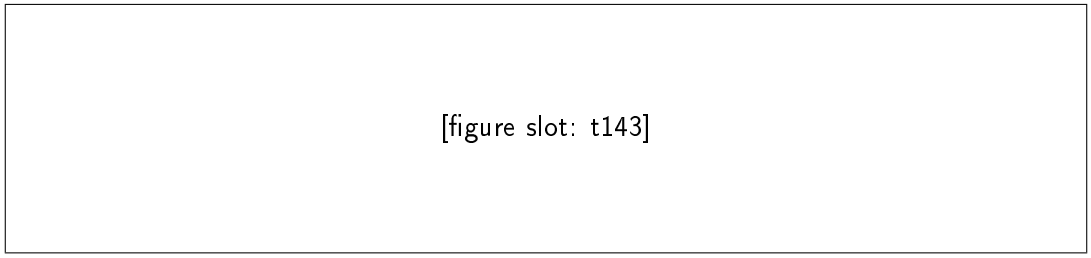


FIGURE 6. The tiling of the triangle with sides $(39, 49, 55)$ by 143 copies of the triangle $(3, 5, 7)$ (angle $2\pi/3$ between the sides 3 and 5), the sporadic count 143 of F_4. It was found by a search in an earlier release and is a Lean theorem there; 143 is not of any of the forms k^2, $x^2 + y^2$, $3k^2$, $6k^2$ and is not in any parametric family of this paper.

6.2. **What is proved about C.**

Theorem 6.2 (The existence half, on the proved ranges; [PROVED IN LEAN]). *Let* ConjCBase(N) *be the statement "$N = k^2 n$, $k \geq 1$, n in the Lean predicate $F_0 \cup \cdots \cup F_6$ with the ranges proved at the time of that module". Then* ConjCBase(N) *implies that N is a tile count.*

Lean: `ConjC.Exist.conjC_existence` (modules `ConjC/Defs`, `ConjC/Exist`, `ConjC/BetaCount`; aggregate `AllConjC`; each family predicate is exactly the parameter range of an existing Lean existence theorem, and the non-vacuity of each is checked on an instance, e.g. `conjCBase_143`, `conjCBase_11625`). It collects constructions under one statement; it is not new mathematics beyond the family theorems it cites. The predicate in Lean is narrower than Table 1: it has F_1 for $M \geq f$ only (Theorem 3.1, the tails and the macro counts of Sections 4–5 are separate Lean theorems, not yet folded into it), it has no F_7, F_8, and its F_5 is the part proved at the time (the row $e = 1$, $g \geq f + 1$; $966g^2$, $g \geq 25$; five engine values).

[COMPUTED, EXACT] *Consistency checks* (exact computations in the working notes, repeated for the census of this release where stated). (a) The families generate exactly the 105 tile counts ≤ 200 of the earlier theorem, no more and no fewer. (b) None of the 199 140 excluded values of the census is generated by any family (rechecked for the Theorem W family and the macro counts in this release: no overlap). (c) Every one of the known tile counts of the census is generated by some family; this check is partly circular, because the known values were found by these families, so it shows that the families are complete for what is known and nothing more. The sharp content of C is the exclusion half.

6.3. **What the exclusion half needs.**

Everything proved in the exclusion direction in the earlier releases is compatible with C. The exclusion half of C is not proved: it is the statement that no value outside the families is a tile count, and it decomposes into threshold statements, one for each family at the place where the known range stops:

- W^* for F_1: $W_M(e,f)$ is tiled iff $M \geq M^*(e,f)$; open for $e > f/2$ (Section 3); and T^*: T_M is tiled iff W_M is.
- V, A' for F_6: the target Ω_g of $\gamma = 2\alpha$ is tiled only if $v \mid g$, and for $g = 1$, $(g-1)(v-u) < u$; open.
- Θ for F_8: $\theta_t(e,f)$ is tiled iff t is in a semigroup, with $te < 2f$ excluded (proved, [PROVED IN LEAN]).
- S_3 for F_5: the row $G(e,f)$ has a smallest element $g_{\min}(e,f)$ and its gaps; known values are $g_{\min}(1,2) = 2$ (exactly, since 21 is not a tile count), $g_{\min}(2,5)$, $g_{\min}(1,6) \in \{1,2\}$, $g_{\min}(1,4) \leq 5$.
- Z for F_3: Zhang's counts m^2ab etc. below the threshold $m < 3t_0$ are not tile counts unless another family covers them; the known exceptions are the $15m^2$ and $33m^2$ counts.
- small leaves: the $\pi/3$ and $2\pi/3$ pairs with irrational angle (closed, [PROVED IN LEAN]), the commensurable cases.

Each of these is an infinite non-existence theorem; the methods of this project (additive invariants and the run induction of the earlier papers) prove them only in the ranges stated, and a count at a single scale cannot decide a threshold statement at all, because the same tile tiles the target at every larger scale. **Expectation.** We expect the true set to be a union of schemas of this type, since a threshold or a semigroup appears in all constructions known; whether the thresholds $g_{\min}(e,f)$ and $M^*(e,f)$ have closed forms for all rows is open (for F_1, W now has one). If a core outside all families is ever found, C needs a further family and the classification becomes an open-ended census. **Falsifiability.** The cheapest tests of C: the smallest value of F_5 outside the known range ($1860 = 465 \cdot 2^2$, 4185, 7440), the squarefree cores 465, 966, 2485 ($g = 1$; C predicts that they are not tile counts), and the first open value of F_6 at $g = v$, namely $1617 = (4, 7, g = 7)$.

7. The census

7.1. **What it counts.** The universe is the 392 074 non-squarefree integers $N \in [2, 10^6]$; N is *excluded* if the decidable hypothesis of the single Lean theorem of the earlier paper fails for it (a program evaluates the hypothesis for each N; the theorem is Lean, the evaluation is computation), *known* if some construction gives a tiling, and *undecided* otherwise. The three sets are disjoint (asserted by the script: no known value is excluded). The baseline is the census of the previous release.

	excluded	known (tile counts)	undecided
version 2.2 / release v1.16	199 140	102 613	90 321
version 3.0 / release v3.0	199 140	129 108	63 826
change	0	+26 495	−26 495

The exclusion side did not change: this release is existence work. The 26 495 new known values, by label:

label	values	source
[PROVED IN LEAN] (formal)	26 312	Theorem 3.1 family (26 138 new), collar and tails, macro counts, 13 W-below closures, k^2-multiples
[COMPUTED, EXACT] only (exact tiling, no Lean)	145	witness-only values from template searches
[PROVED BY HAND] (hand)	35	collar step with $h > 1$ on the rows $(1,f), (2,3), (2,5)$ (Theorem 4.1)
paper-formal closure over exact seeds	3	upward closure statement of the earlier paper over verified seeds

(The 102 613 known values of the baseline keep the labels of the previous release.) Theorem W alone moves 26 138 values from undecided to known: the family has 44 495 values $\le 10^6$ (28 946 from W, 18 431 from T, union 44 495; the largest f is 251); 18 357 of them were already known by other means, none lies in the excluded set. **Paliy-claim-only values.** A further 134 values $\le 10^6$ lie in families that Paliy claims ([CLAIMED BY PALIY]; the first are 1380, 2024, 3105, 4554, 5520, 8096) and that we have neither proved nor verified; they are *not* counted as known, they are among the 63 826 undecided.

7.2. **Squarefree values.** Of the squarefree $N \le 10^6$, 770 remain open at the exact level (771 in version 2.2: the difference is 4830, a tile count by Paliy's certificate, exact-verified here); at the Lean level the count of the previous release (791) is unchanged, since all new Lean theorems concern non-squarefree values. The open squarefree values ≤ 1000 are 231, 329, 354, 385, 455, 465, 517, 777, 806, 935, 966. The undecided non-squarefree values in $[201, 1000]$ are now 38: 236, 270, 280, 284, 297, 332, 351, 412, 420, 423, 444, 480, 524, 531, 552, 594, 616, 644, 668, 716, 732, 747, 750, 816, 824, 840, 855, 876, 880, 908, 912, 924, 927, 944, 945, 956, 963, 999 (414 and 639, tile counts of $W_3(2,5)$ and $T_3(2,5)$, are no longer on the list). Every $N \le 200$ was already decided.

7.3. **What the census is not.** It is not a classification: the undecided values may be tile counts or not. It rests on the families known to us; a tile count in a family we do not know is invisible to it. The excluded values are excluded by Lean theorems, but the assignment "this N satisfies the hypothesis" is made by a program, not by the kernel (as in the earlier paper). The known values are of the stated labels; only the [PROVED IN LEAN] ones are Lean theorems about *that* N.

8. Computer evidence on one open family: the W^{3+} census

This section records exact computations that are *not* proofs and *not* formal.

8.1. **The family and the question.** Let $u < v < 2u$ be coprime, $b = v^2 - u^2$, tile (u^2, b, uv) (this is the family F_6, with $\gamma = 2\alpha$), and Ω_g the isosceles target with base angles α, legs gub, base gvb; a tiling has $g^2 b$ tiles. The earlier papers prove that Ω_g is not tiled when $(g-1)(v-u) < u$ (Theorem F) and conjecture that it is tiled only if $v \mid g$ (V) and $g \ge g_D(u,v)$ (A'); the only tilings known are three Herdt-type counts, 1125, 19 600, 130 977. The first open instances with $v \nmid g$ and $(g-1)(v-u) \ge u$ are $g = 3$ for $(u,v) = (4,7)$ (tile $(16, 33, 28)$, $N = 297$) and $(5,8)$ (tile $(25, 39, 40)$, $N = 351$).

8.2. **The words.** In a tiling of Ω_g, the tiles with an edge on a leg partition the leg into segments; reading from the corner α to the apex gives a *word*: the sequence of edge letters (a, b or c) with the angles of the tiles at their ends. The length of the leg and the angle sums at the ends leave a finite list of admissible words; hand-proved filters (called D_2, D_1, K_2, A', G_1 in the working notes) and a rule for the junction at the apex cut it down to the set we call W^{3+}. For each word an exhaustive exact search (integer arithmetic, no floating point) asks whether the part of a tiling near that leg exists. The verdict "no tiling" for a word is therefore conditional on the filters; it covers every tiling whose leg word is that word, and *only* such tilings (it is not a verdict on Ω_g unless it is returned for every word, including the words outside W^{3+} that the filters exclude).

(u,v), $g=3$	words	no tiling	tiling found	open
$(4,7)$	191 940 (complete list)	191 940	0	0
$(5,8)$	84 462 (complete list)	84 462	0	0
$(7,12)$	8 431 936 799	8 014 982 151 (95.06%)	0	416 954 648

For $(4,7)$ and $(5,8)$: all but 323 and 630 words fall within 2 s each; the remainder were rerun at 120 s, 1800 s, and without a time limit (the two hardest words of $(4,7)$ took 5614 s and 3269 s); every word is decided. For $(7,12)$ the word list has 8.4×10^9 entries and was covered by an adaptive prefix tree (a "no tiling" for a prefix refutes all its completions: the justification is that no tile edge straddles the end of a word prefix); the residual 4.2×10^8 words are open and the run was stopped because the fraction of prefixes that time out grew with depth. **Controls.** The search finds the known tilings when it is given their own words (the words of the tilings of the scaled tile at $m = 3, 4$, for all three pairs: found and verified), and refutes the reversed words and words refuted independently by a slower program. **What this shows and does not show.** No tiling was found, so V and A' are not refuted. It does not show that 297 and 351 are not tile counts, because the verdict is for the one-leg word set, not for all tilings and all shapes of target; and without a Lean certificate it is not a theorem. It is evidence, and it narrows where a counterexample could be.

9. Corrections

Statements made in our working notes, and in one case in an earlier release, that this paper corrects. None of the corrections affects a Lean theorem, a prime-case statement or an excluded value of the census.

(1) **183 is not a tile count.** An earlier draft of the conjecture listed 183 as a sporadic tile count (attributed to Laczkovich). 183 is excluded, and every $N \le 200$ is decided in Lean; the draft misread a statement about $5 \cdot 183^2$. The Lean predicate F_4 never contained it.

(2) **Tiles of T_M and W_M below the scale f exist for $e \ge 2$.** Our notes asserted that $T_M(e,f)$ and $W_M(e,f)$ have no tiling for $M < f$ when $e \ge 2$ ("B_1"), and that the T half of this statement stood after the W half failed. Both are false: Theorem 3.1 and the glue of Section 3.3. The version 1 paper [11] recorded, as an observation, that "in every case decided so far" the scale f is the threshold; Corollary 3.3 is a counterexample ($W_4(3,5)$, $M = 4 < 5$). The open problems of that paper on the scales $f/(2e) < M < f$ are partly closed here (Section 3).

(3) **The template "exists iff $4f < 9e$" is false.** An intermediate note (13 witnesses below f, then a replay of one search) stated that the six-piece template exists only for $4f < 9e$, was absent for 432 coprime pairs with $f \le 60$, and that the conjectured threshold held only there. This was an artefact: the replay of one search history has an empty cone for the other pairs, but the geometry of the same six pieces works for all $2 \le e \le f - 2$ (Theorem 3.1; for $(2,5)$, $(3,7)$, $(4,9)$, $(5,12)$, $(3,8)$ the tilings exist).

(4) **The size conjecture is false.** The working notes (and a conjecture in a longer draft) stated that a target not similar to the tile has longest side at least $c^2 = f^4$. $W_4(3,5)$ has longest side $500 < 625$.

(5) **Paliy's 4830 is a tiling.** Our notes said that his 4830-tile file is not a tiling. We had checked the wrong file (his refuted moment counterexample, not his certificate). His certificate, 4830 copies of $(24, 11, 31)$ on the triangle $(961, 1155, 1426)$, is verified exactly (Section 5).

(6) **The θ-isosceles statement.** Our notes stated "targets $\theta_t(e,f)$ with $te \ge f$ have no tiling". That is false: $\theta_4(1,2)$ ($te = 8$) and $\theta_8(2,3)$ are tiled. The proved statement is the necessary bound $te \ge 2f$; the true statement is a threshold (a semigroup of scales, Paliy), not an exclusion.

(7) **Priority of the α up-set.** The up-set in g of the base-α rows and the form $b\kappa K^2$ are Paliy's, dated 29 September 2026 (Section 4); our notes had presented the up-set as ours. The scale additivity of the rows is also his.

(8) **Counts in a note.** The census effect of the collar step was first stated as 235 values; the correct figure, by an audit that recounted, was 194 (119 in the rows $(1, f), (2, 5)$ and 75 in $(2, 3)$); the figures of this release supersede both.
(9) **The threshold formula** $(M-1)(f-e) \geq f$**.** Used as a threshold for all rows it contradicts the proved side theorem W_M none if $Me < f$ when $e/f < (3-\sqrt{5})/2$ (for instance $(2, 7)$: the formula gives $M = 3$, but $3 \cdot 2 < 7$). The correct threshold is $M^*(e, f) = \max(\lceil f/e \rceil, \lceil f/(f-e) \rceil + 1)$.
(10) **The pattern** $g_{\min} = f + 1$**.** Our notes observed that the first realised scale of a base-α row was $f + 1$ for even f and recorded this as a pattern; $g_{\min}(2, 5)$ and $g_{\min}(1, 6)$ are ≤ 2 (Theorem 4.3).

The register of corrections of the project is in the repository (file `TIMELINE.md` records the dates).

10. What is not proved

- **The exclusion half of Conjecture C.** No description of the set of tile counts above 200 is proved; the census has 63 826 undecided non-squarefree values and 770 squarefree values $\leq 10^6$.
- **The no side of Theorem W for** $e > f/2$**:** for each row exactly the scales M with $(M-1)(f-e) < f < (M+1)(f-e)$ (at most two) are undecided; and $e = 1$ **and** $e = f - 1$ **for** T_M (first case $T_3(1, 4)$); the T side in general (only $2Me < f$ is excluded).
- V **and** A' **for** $\gamma = 2\alpha$**:** the evidence of Section 8 is not a proof; $(7, 12)$ is incomplete (95.06%).
- Θ **and** S_3**:** the thresholds of the θ-isosceles family and of the base-α rows; the squarefree cores 465, 966, 2485; the values 1860, 4185, 7440.
- $15m^2$ **for every** $m \geq 4$ **as a Lean theorem**; the collar step for arbitrary height h (hand only); Paliy's claims that we have not verified (134 values $\leq 10^6$ in the census; the mixed-corner family beyond 4830).
- **The squarefree values:** 770 open; the squarefree tile counts known to us are of the classical forms, or come from a short list of sporadic and layout values (the 22 of the earlier release, 77, 143, 4830).
- **Outside check.** The Lean statements and the proofs have been compared and audited only inside this project, including the statement of `theoremP_6` (the prime case of the earlier releases) and of Theorem W; a check by a reader outside the project has not been made, and announcing the prime case as proved remains the author's decision.
- The search that found the macro layouts is not part of this release and is not described further than in Section 5; its outputs are verified independently.

11. Credits, with dates

"Ours": statement and proof produced in this work. "Formalized": another author's statement and construction, our Lean proof. Dates are of first public record (repository commit dates, arXiv versions) unless marked.

result	status here	whose, and when
Prime counts; the base-β reduction	earlier releases	Paliy, first public proof 2026-09-29 [9]; ours independent; exclusion of $e = 1$ public 2026-09-12 (Zenodo v3.2); Beeson's withdrawal note for arXiv:2607.19572 (v2, 2026-09-24) [5] credits the author, cited here as public record
Targets W_M, T_M; tilings for $M \geq f$	formalized	Paliy (W/β framework, caps, collars: 2026-10-03; all $M \geq v$: 2026-10-10); Beeson and Roşu (the targets, the seed)
Theorem W, six-piece template	ours	template found by our search; the setting and the earlier thresholds are Paliy's and Beeson's; the sharpness side ($Me < f$) is from the earlier releases
α up-set, form $b\kappa K^2$, scale additivity	Paliy's; Lean here	Paliy, 2026-09-29 [10]; seeds 84, 189: Beeson (Figs. 28–29), 189: Harries
Collar step, parametric seed $e = 1$, tails	ours	independent of Paliy's construction; credit for the up-set as above
Macro pieces	ours (layouts, Lean)	macro-dissections: Laczkovich; Herdt (1125); seeds $15m^2$: Paliy (2026-10-08)
$15m^2$ for all $m \geq 4$, 4830, mixed-corner family	Paliy's ([CLAIMED BY PALIY]); 4830 verified here	Paliy, 2026-10-08
990, 264, 897, 240, 312	earlier releases	Paliy: 990 on 2026-10-06, 264 on 2026-10-07; the others as dated in the earlier releases
θ-isosceles family, scale additivity	Paliy's ([CLAIMED BY PALIY])	Paliy, 2026-09-29
Run induction, Theorem F and D'	earlier releases	method: Beeson and Roşu (preprint of 2026-10-16); $g = 1$: Beeson 2012
1125-tiling	earlier releases	Herdt, published by Beeson
88	earlier releases	Harries
$2\pi/3$ families F_3, $15m^2, 33m^2$	formalized	Zhang (arXiv:2512.22696); seeds Paliy, Herdt
Rep-tiling and F_0	—	Snover, Waiveris and Williams
Classification of pairs (triangle, tile)	—	Laczkovich
Conjecture C, census, W^{3+} evidence	ours	—

The statements "Paliy claims" in this paper are statements in his public repository [10]; we have read his documents and checked some examples, and we say so where we did. Nothing here is a claim of priority over him for the results listed as his; the dates are given so that the order is visible.

Appendix A. Proof of Theorem W

Credit: the targets W_M, T_M and the framework are Beeson's and Paliy's; the six-piece template was found by our search, and every identity below was also confirmed by a computer algebra system and by exact integer computation; Lean checks the whole argument (Section 3). Throughout, e, f are coprime, $2 \le e \le f-2$, $g = f-e \ge 2$, and a, b, c, κ are as in Section 2.1.

A.1. **Complex coordinates.** Put $w = \sqrt{4f^2-e^2}$ and write the complex number $x+iwy$ as $[x,y]$. Then $\mathrm{Area}(\mathsf{t}) = ebw/4$, and

$$\varepsilon = e^{i\alpha} = \frac{\kappa + iew}{2f^2}, \qquad \eta = e^{i\beta} = \frac{e(3f^2-e^2)+ibw}{2f^3}, \qquad \omega = \overline{\varepsilon}\overline{\eta} = e^{-i(\alpha+\beta)}.$$

Polynomial identities (clear denominators and expand): $\varepsilon^3\eta^2 = -1$ (this is $3\alpha+2\beta=\pi$); $a + b\omega = c\bar{\eta}$ (the tile closes up); $f\omega + e\varepsilon = f\bar{\eta}$. A triangle similar to t with sides sa, sb, sc is a union of s^2 tiles, and a parallelogram with sides sa, uc and angle β is a union of $2su$ tiles (Proposition 2.1).

A.2. **The template.** Let M, t, r, q be positive integers and put

$$v = fM - er, \quad s_1 = f(r-t), \quad s_2 = fr - eM, \quad x = ft - eq, \quad u_1 = fq - et, \quad u_2 = fq - er.$$

Points (vector sums in $\mathbb{C}$):

$$\begin{aligned}
&O = 0, && B = fbM, && A = f^3M\varepsilon^2, && P = fbM\varepsilon,\\
&P_2 = b(fM-et)\varepsilon, && P_3 = P - eat\omega, && Q = P + fbt\omega, && R = P + fbr\omega,\\
&C' = bv\varepsilon, && B' = cv, && V_2 = P + f^3t\bar{\eta} - ef^2r, && X_4 = P - fbq\varepsilon,\\
&U_1 = P - ef^2t + ef^2q\bar{\eta}, && U_2 = P - ef^2r + ef^2q\bar{\eta}.
\end{aligned}$$

The target is the triangle OAB: $|OB| = Mfb$, $|OA| = Mf^3$, $|AB| = Me\kappa$, with angles 2α at O, β at A, $\alpha+\beta$ at B; that is $W_M(e,f)$. The point P lies on AB ($A-P = -afM\omega$, $B-P = ebM\omega$) and $OP = OB$. The pieces, counter-clockwise:

	vertices	kind	description	tiles
π_0	A, O, P_2, P_3	trapezoid	triangle O, A, P (scale fM) minus the corner triangle P, P_2, P_3 (scale et)	$f^2M^2 - e^2t^2$
π_1	O, B', C'	triangle	scale v	v^2
π_2	P_3, P_2, Q	triangle	scale ft	f^2t^2
π_3	B, Q, V_2, B'	trapezoid	triangle R, Q, V_2 (scale s_1) minus the corner R, B, B' (scale s_2)	$s_1^2 - s_2^2$
π_4	Q, U_1, U_2, V_2	parallelogram	sides as_1 ($V_2 \to Q$) and cx ($Q \to U_1$)	$2s_1x$
π_5	U_1, P_2, C', U_2	trapezoid	triangle X_4, U_1, P_2 (scale u_1) minus the corner X_4, C', U_2 (scale u_2)	$u_1^2 - u_2^2$

The area identity $\sum(\text{tiles}) = M^2(2f^2-e^2)$ is an identity in $\mathbb{Z}[e,f,M,t,r,q]$ (checked by computer algebra).

A.3. **Lemma A (shapes).** *If $s_1 > s_2 > 0$, $u_1 > u_2 > 0$, $v, ft > 0$ and $fM > et$, each piece is the stated union of tiles.*

- π_0: $OP = bfM\varepsilon$ and $OA = cfM\varepsilon^2$ have lengths bfM, cfM enclosing α: SAS gives a triangle of scale fM. P_2 lies on OP and P_3 on PA with $PP_2 : PO = PP_3 : PA = et : fM$, so $P_2P_3 \parallel OA$ and the cut-off triangle PP_2P_3 has scale et.
- π_1: $OB' = cv$ and $OC' = bv\varepsilon$ enclose α: scale v.

- π_2: $P_3Q = (ea+fb)t\omega = f^3t\omega$ has length $c\cdot ft$, and $|P_3P_2| = c\cdot et = a\cdot ft$; the angle at P_3 is the angle β of the triangle PP_2P_3, similar to OPA: scale ft.
- π_3: $Q-V_2 = as_1$ is real (this is the closing identity $a+b\omega = c\bar{\eta}$ times ft); $R-Q = -bs_1\omega$; the angle at Q between the directions ω and π is γ: sides a, b enclosing γ, scale s_1. $B = R - bs_2\omega$ lies on RQ and B' lies on RV_2 with $RB' : RV_2 = s_2 : s_1$.
- π_4: $U_1 - Q = -f^2x\bar{\eta} = U_2 - V_2$, so $V_2QU_1U_2$ is a parallelogram with sides as_1, cx and angle β at Q.
- π_5: $P_2 - X_4 = bu_1\varepsilon$; $|U_1 - X_4| = cu_1$, $|U_1 - P_2| = au_1$; C' lies on X_4P_2 with $X_4C' : X_4P_2 = u_2 : u_1$, and U_2 on X_4U_1 with the same ratio.

Each statement is a polynomial identity in e, f, M, t, r, q (clear denominators) and was checked by computer algebra; the Lean proof has one lemma for each. □

A.4. **Lemma B (geometry reduces to inequalities).** All six pieces are convex and counterclockwise when the scales are positive (their doubled signed areas are the tile counts times a positive constant). Hence a piece lies in the target iff its vertices do, and two pieces have disjoint interiors if one edge line of one has the other piece on its closed outer side; the edges needed are: $\pi_0|\pi_1$: OP_2; $\pi_0|\pi_2$: P_2P_3; $\pi_0|\{\pi_3,\pi_4,\pi_5\}$: OP_2; $\pi_1|*$: $B'C'$; $\pi_2|*$: P_2Q; $\pi_3|\{\pi_4,\pi_5\}$: QV_2; $\pi_4|\pi_5$: U_1U_2. With the area identity this gives a tiling (closed convex pieces inside the closed convex target of total area equal to its area cover it). Every cross product needed is a product of positive constants (depending on e, f) and factors from the list below; the sign of the constant prefactors is read once, at one instance, and is independent of the parameters:

- C2: $r - t > 0$; C3: $eM - ft > 0$; C4: $fr - eM > 0$; C5: $fM - er > 0$;
- C6: $fq - er > 0$; C7: $ft - eq > 0$; C8: $f^2t - e^2r > 0$; also $M > q$, $t > 0$, $r > 0$;
- LA: $bfM - e\kappa t + f^3q > 0$; LB: $bfM - e\kappa r + f^3q > 0$;
- LC: $efbM - e^2\kappa r + f^4t > 0$

(together with three expressions that are positive for any positive parameters: $bM+eft$, $eMf+bt$, $br + e^2t$; LB implies LA since $r > t$). *Lean:* one lemma per factor (modules `ThmWIneq0`–`ThmWIneq4`).

A.5. **Lemma C (the parameters satisfy the list).** Assume (H1), (H2). Put $n = \lfloor eM/f \rfloor$, $\rho = eM - fn \in [0, f)$, $s = M - n$, $\delta = 1$ if $\rho = 0$ and $\delta = 0$ otherwise, and

$$t = n - \delta, \qquad r = n + 1, \qquad q = n.$$

Facts. (F1) $gM = fs - \rho$, and (H2) gives $gM \geq f + g$; so $gM/f > 1$ and $s = \lceil gM/f \rceil \geq 2$. (F2) $n \geq 1$ by (H1). (F3) If $\rho = 0$ then $f \mid M$ (coprimality), $M = fm'$, $n = em' \geq e \geq 2$. (F4) $eg \geq f+1$: $eg - f = (e-1)(g-1) - 1 \geq 1$, since $e, g \geq 2$ and not both equal to 2 (coprime). (F5) $\boldsymbol{gn \geq e+1}$. If $g \geq e+1$: $gn \geq g \geq e+1$. If $g \leq e$: write $f = gh + j$ with $0 < j < g$ (coprimality of g and f) and $h \geq 2$; then $e = g(h-1) + j$, and (H2) gives $M - 1 \geq \lceil f/g \rceil = h + 1$; now $e(h+2) - hf = g(h-2) + 2j > 0$, so $eM > hf$, $n \geq h$, and $gn \geq gh = f - j \geq f - g + 1 = e + 1$.
The factors.

- $t \geq 1$: for $\delta = 0$, $t = n \geq 1$; for $\delta = 1$, $t = n - 1 \geq e - 1 \geq 1$ by (F3). And $r > t$, $r > 0$, $q = n > 0$. $M > q$ because $n \leq eM/f < M$. (C2 holds.)
- C3: $ft \leq fn \leq eM$, with equality only if $\delta = 0$ and $\rho = 0$, which is impossible; for $\delta = 1$, $ft = fn - f < eM$.
- C4: $fr - eM = f - \rho > 0$.
- C5: $fM - e(n+1) \geq (bM - ef)/f > 0$ because $bM = (f+e)gM \geq (f+e)(2f-e) = 2f^2 + ef - e^2 > ef$ (by F1: $gM \geq f + g = 2f - e$).
- C6: $fq - er = gn - e \geq 1$ by (F5).

- C7: for $\delta=0$, $ft-eq=gn>0$; for $\delta=1$, $ft-eq=f(n-1)-en=gn-f\geq ge-f\geq 1$ by (F3), (F4).
- C8: for $\delta=0$, $f^2t-e^2r=bn-e^2=(f+e)gn-e^2\geq(f+e)(e+1)-e^2>0$; for $\delta=1$, it is $bn-f^2-e^2\geq be-f^2-e^2=eg(f+e)-f^2-e^2\geq(f+1)(f+e)-f^2-e^2>0$.
- LB: substituting $M=(fs-\rho)/g$ and $n=(eM-\rho)/f$ one finds the polynomial identity $\mathrm{LB}=s\Psi-\kappa(\rho+e)$ with $\Psi=f^3+2ef^2-e^2f-e^3>0$ (checked by computer algebra). With $s\geq 2$ and $\rho\leq f-1$: $\mathrm{LB}\geq 2\Psi-\kappa(f+e)+\kappa>2\Psi-\kappa(f+e)=e(2f+e)(f-e)>0$. Then $\mathrm{LA}=\mathrm{LB}+e\kappa(r-t)>0$.
- LC: $\mathrm{LC}=efbM+b^2n-e^2\kappa-\delta f^4$ (using $f^4-e^2\kappa=b^2$). For $\delta=0$: $efbM=ef(f+e)gM\geq ef(f+e)(2f-e)$, so $\mathrm{LC}\geq e\,[f^2(2f-e)-e^2(f-e)]>0$. For $\delta=1$: $M=fm'$, $n=em'$, $\mathrm{LC}=m'eb\kappa-e^2\kappa-f^4\geq e\kappa(b-e)-f^4$, and $e(b-e)=eg(f+e)-e^2\geq 2(f-2)(f+2)-(f-2)^2=f^2+4f-12>f^2$ (using $eg\geq 2(f-2)$ for $2\leq e\leq f-2$, $f+e\geq f+2$, $e\leq f-2$), while $\kappa>f^2$; so $\mathrm{LC}>0$.

Positivity of the scales: $v=fM-er>0$ is C5, $s_1=f(r-t)>0$ is C2 times f, $s_2=fr-eM>0$ is C4, $s_1>s_2$ is $f(r-t)>fr-eM$, i.e. C3, $x=ft-eq>0$ is C7, $u_1=fq-et>0$ follows from C6, $u_1>u_2$ is e times C2, $u_2=fq-er>0$ is C6, and $fM>et$ follows from C3. Lemmas A, B and C prove Theorem 3.1 for W_M; the count of tiles is $M^2(2f^2-e^2)$ by the area identity. The inequality $4f<9e$ is not used anywhere. □

A.6. **Evidence that the proof is right.** See Section 3.6. In addition, the parameter triple (t,r,q) is not unique (for instance $q=n+3$ also works at $(7,13,13)$), and the canonical triple satisfies the whole list exactly when $M\geq M^*(e,f)$, for all coprime pairs with $f\leq 40$ and $M<4f$: the bound (H1)+(H2) is exactly the range in which the canonical parameters work.

Appendix B. Data: the explicit tilings

B.1. **The thirteen first witnesses with $M<f$.** Found first by a computer search, and now instances of Theorem 3.1; every one is a Lean theorem (module `Draft/DebtBWTilable`, with the closure upward in M).

(e,f,M)	W_M tiles	T_M tiles	(e,f,M)	W_M tiles	T_M tiles
$(3,5,4)$	656	1056	$(7,12,4)$	3824	6128
$(4,7,4)$	1312	2096	$(7,10,5)$	3775	6275
$(5,8,4)$	1648	2672	$(7,9,6)$	4068	6984
$(5,7,5)$	1825	3050	$(8,11,5)$	4450	7475
$(5,11,3)$	1953	3042	$(9,11,7)$	7889	13 818
$(5,9,4)$	2192	3488	$(6,11,4)$	3296	5232
$(7,11,4)$	3088	5024			

B.2. **The shipped files.** Directory `data/witnesses/v3_exact`: 97 files, each named by the number of tiles, the kind and the parameters (for example `tiling_414_W_2_5_M3.exact.txt.gz`). Each begins with a comment line stating the number of tiles and the side lengths of the tile; every other line gives the three vertices of one tile in exact rational coordinates $(X,Y\sqrt{D})$. Kinds: `W`, `T` (Theorem W and its glue: 37 and 15 files), `collar` (6) and `macro` (38) (Sections 4, 5), `F3_paliy` (Paliy's 4830). Verification: the exact verifier of the repository (`bte verify`) reports, for each file, that the tiles are congruent to the stated tile, that their interiors are pairwise disjoint, that the total area equals the area of the hull, and that the hull is a triangle. All 97 pass.

Appendix C. Index of the Lean statements

All in the library of the repository (directory `lean/`); aggregate `Erdos634.AllV3` (the modules new in this release; build one module at a time through `code/lakeserial.py`); the axioms of 98 of them are printed by `lean/axioms/AxiomsAllV3.lean` and every line reads `[propext, Classical.choice, Quot.sound]`.

statement	Lean name (namespace `Erdos634....`)
Theorem 3.1	`Draft.ThmW.thmW`; counts `Draft.ThmW.tileCount_W`, `tileCount_T`; examples `inst_414`, `inst_656`, `inst_639`, `inst_27324`
13 first witnesses, closures	`Draft.DebtB.closure_656` etc.; `DebtB.counts_of_wTilable`
Collar layer step	`Draft.CollarLayer.layer_core`, `Draft.CollarFam.layer_step`
Theorem 4.2	`Draft.CollarUp.tileCount_e1_up`; instances `tileCount_189`, `tileCount_16740`, `tileCount_159040`
Theorem 4.3	`Draft.CollarFam.tileCount_465_sq`, `Draft.CollarFam5.tileCount_2485_sq5`, `Draft.CollarFam17.tileCount_966_sq17`
Macro pieces (Thm 5.1)	`Draft.MacroPiece.macroTFam_gen`; `Draft.TM375.tileCount_375` etc.; `Draft.T62125.tileCount_62125`; `Draft.AllMacro.tileCount_78246`
Twelve square multiples	`Draft.TemplateSquares.tileCount_46500` etc.
Conjecture C, existence half	`ConjC.Exist.conjC_existence`; `ConjC.BetaCount.tileCount_beta_ge`

Appendix D. Reproducing the checks

From a clean clone, with the toolchain of `lean/lean-toolchain`: build the aggregate one module at a time, print the axioms, and verify the exact tilings:

```
python3 code/lakeserial.py lean build Erdos634.AllV3
(cd lean && lake env lean axioms/AxiomsAllV3.lean) \
   | python3 code/axiom_gate.py
BTE=rust/bonfioli_tiling_engine/target/release/bte
for f in data/witnesses/v3_exact/*.gz; do
  $BTE verify "$f" | grep -q VERIFIED || echo FAIL "$f"
done
```

The build needs about 6 GB of memory for its largest module and about 22 GB of disk for the build products; one module is built at a time. The scripts `code/reproduce.sh` (all earlier checks) and the above give the complete set.

References

[1] M. Beeson, *Triangle tiling I: The tile is similar to ABC or has a right angle*, arXiv:1206.2231.
[2] M. Beeson and G. Roşu, *Tiling a triangle into a prime number of congruent triangles*, preprint, version of 16 October 2026.
[3] M. Beeson, *Triangle tiling III: the case* $3\alpha + 2\beta = \pi$, arXiv:1206.2229.
[4] M. Beeson, *Tilings of an isosceles triangle*, arXiv:1206.1974.
[5] M. Beeson, *No prime tiling of an isosceles triangle*, arXiv:2607.19572 (withdrawn in version 2, 24 September 2026, with a note attributing the theorem to V. Bonfioli).
[6] M. Beeson, M. Laczkovich and Y. X. Zhang, *Solution of Erdős Problem 633*, arXiv:2604.03609.

[7] M. Laczkovich, *Tilings of triangles*, Discrete Math. **140** (1995), 79–94.
[8] M. Laczkovich, *Tilings of convex polygons with congruent triangles*, Discrete Comput. Geom. **48** (2012), 330–372.
[9] D. Paliy, *No scale-one triangle in either rational Group-1 family*, and the deduction of the prime case, repository `github.com/DenisUkranian/erdos-634-tilings`, files `docs/prime-case-candidate.md` and `docs/prime-case-dependencies.md`, first published 29 September 2026; unrefereed.
[10] D. Paliy, repository `github.com/DenisUkranian/erdos-634-tilings`: notes on eventual rational families (`docs/eventual-rational-families.md`), the θ-branch (`docs/theta-branch.md`), the W/β framework and the macro-dissection files, September–October 2026; unrefereed.
[11] V. Bonfioli, *Congruent dissections of a triangle: prime counts, the counts up to* 200*, and a family of isosceles targets*, release v1.13.1, Zenodo, October 2026, doi:10.5281/zenodo.23400433.
[12] V. Bonfioli, *Congruent dissections of a triangle: runs, windows and the corner*, version 2.2, release v1.16, Zenodo (all versions), doi:10.5281/zenodo.23400433.
[13] S. L. Snover, C. Waiveris and J. K. Williams, *Rep-tiling for triangles*, Discrete Math. **91** (1991), 193–200.
[14] A. Soifer, *How does one cut a triangle?*, 2nd ed., Springer, 2009.
[15] Y. X. Zhang, *Tiling triangles with* $2\pi/3$ *angles*, arXiv:2512.22696.